KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2017
Department of Computer Science and Engineering
CSE 4207
Computer Graphics

TIME: 3 hours FULL MARKS: 210

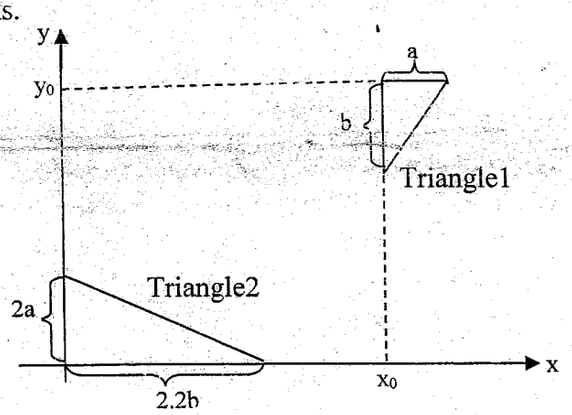
N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

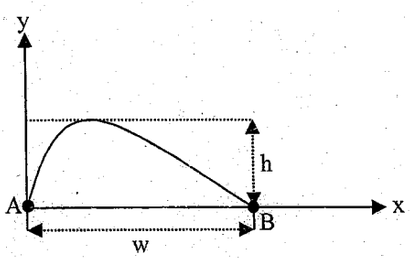
1. a) Explain the differences between "object coordinates", "world coordinates", and "camera coordinates". (08)
   b) When can you convert a quadratic Beizer curve to a cubic Hermite curve? (07)
   c) In Breshenham's algorithm, we saw the efficiency of incremental evaluation. For the function, $f(x) = y = x^2 - 3$, compute the incremental evaluation function. (10)
   d) If $(x_0, y_0) = (5,4)$ and $(x_{end}, y_{end}) = (8,10)$, use Breshenham's algorithm to compute pixels for the line segment. (10)

2. a) Prove that $R(\theta_1).R(\theta_2) = R(\theta_1 + \theta_2)$ where $R()$ is a rotation function and θ_1, θ_2 are angle of rotation. (07)
   b) Transform Triangle1 to Triangle2 in the following figure by a sequence of basic transformations. Write the transformation matrices in 2D homogeneous coordinates. Compute the final transformation matrix. Note that size of the triangle has also been changed; see the figure for the dimensions. (14)



   c) Reflect the diamond-shaped polygon whose vertices $A(-1,0), B(0,-2), C(1,0)$ and $D(0,2)$ about the line $y = x + 2$. (14)

3. a) We want to draw a cubic Beizer curve starting from $A = (0,0)$ to the point $B = (w,0)$. At A the curve is tangent with y-axis and at B the curve is tangent with x-axis. (see figure below). Draw control points and give the possible positions of them (ignoring h), use new parameters where necessary. (10)



   b) Define fractal. What are the classifications of fractal objects? How is fractal generated? (09)
   c) Find fractal dimension of Koch curve. (05)
   d) Explain depth-buffer algorithm in details for visible surface detection. What are the advantages and disadvantages of z-buffer algorithm? (11)

4. a) How is painting priority of a surface determined by painter's algorithm? (09)

b) A Beizer curve is to be drawn given the control points $P_1(40,40), P_2(10,40), P_3(60,60), P_4(60,0)$. Calculate the coordinates of the points on the curve corresponding to the parameter $t = 0.2, 0.4, 0.6$. Draw a rough sketch of the curve and show coordinates of the various points on it. (14)

c) What is the advantage of using curved surfaces for rendering? How to display it efficiently in practice? (07)

d) What are the drawbacks of Digital Differential Analyzer (DDA) algorithm? (05)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What are the three stages of computer graphics pipe-line? Briefly explain each of them. (10)

b) Let a line L, defined in parametric form below. Determine a rotation matrix about L with rotation angle $= 30°$. (14)

$$L: \begin{cases} x = 2 + 3t \\ y = 4t \\ z = 3 + 4t \end{cases} \qquad 0 \le t \le 1$$

c) When performing Cohen-Sutherland line clipping, how do we use outcodes to check for trivial rejection (what exact operation and comparison would you use?) (06)

d) What is GPU? Can it be used in Phong shading? Why? (05)

6. a) Explain the importance of projection in computer graphics. Find a parallel projection matrix with DOP $V = aI + bJ + cK$ and xy as projection plane. (10)

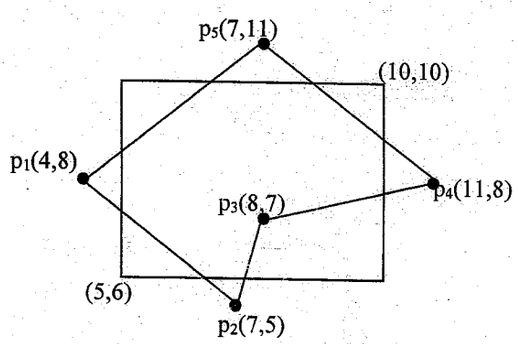
b) Consider a point $P = (2,3,4)$ in eye space. Let a symmetric projection window is defined with aspect ratio $= 1.25$, fovy $= 60°$, near and far plane of the view volume is at a distance 1 and 5 respectively. Find the perspective transformation of the point P in normalized device coordinate space. (12)

c) What is the importance of maintaining pseudo depth in perspective transformation? (05)

d) Describe the anomalies of perspective projection. (08)

7. a) What is view volume? Describe how a parallelepiped view volume can be converted to canonical view volume. (10)

b) Consider the polygon defined by the points p_1, p_2, p_3, p_4, and p_5 as shown in figure below. Use Sutherland-Hodgeman algorithm to clip polygon against the rectangular window defined by the point $(x_{min}, y_{min}) = (5,6)$ and $(x_{max}, y_{max}) = (10,10)$. Compute the final points list. (15)



c) In ray tracing a large computational cost is associated with determining ray-object intersections. Suggest a method that may reduce this cost. (10)

8. a) The normal at a vertex $(0,1,2)$ on a surface is $(0,4,5)$. The light source is at $(0,1,4)$. The diffuse color of the light is $(0.9,0,0.2)$. The diffuse reflectivity of the surface are $(0.4,1,0)$. What will the RGB color of the vertex be? Assume there is no specular, emmisive or ambient component or light attenuation. (10)

b) Explain how light intensity attenuation can be modeled. (07)

c) Let S, be a sphere of radius 10 centered at $(10,-2,-5)$. Determine if a ray $S = 2J + 5K$ and $d = I - 2K$ intersects the sphere. (10)

d) Compare the working principle of Phong and Gouraud shading. (08)